A CENSUS OF INDIA'S PUBLIC WEB

12,092 government portals, measured from inside the country.

The first census-scale observatory of India's public digital infrastructure, audited from domestic peering transit against WCAG 2.1 AA, Core Web Vitals, CERT-In hardening and GIGW 3.0.

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DOMAINS	REACHABLE	MEAN ACCESSIBILITY	MEAN SECURITY
12,092	10,247 · 84.7%	93.8 / 100	68.0 / 100
Seven governance tiers	Measured in-country	WCAG 2.1 AA structural	CERT-In header hardening

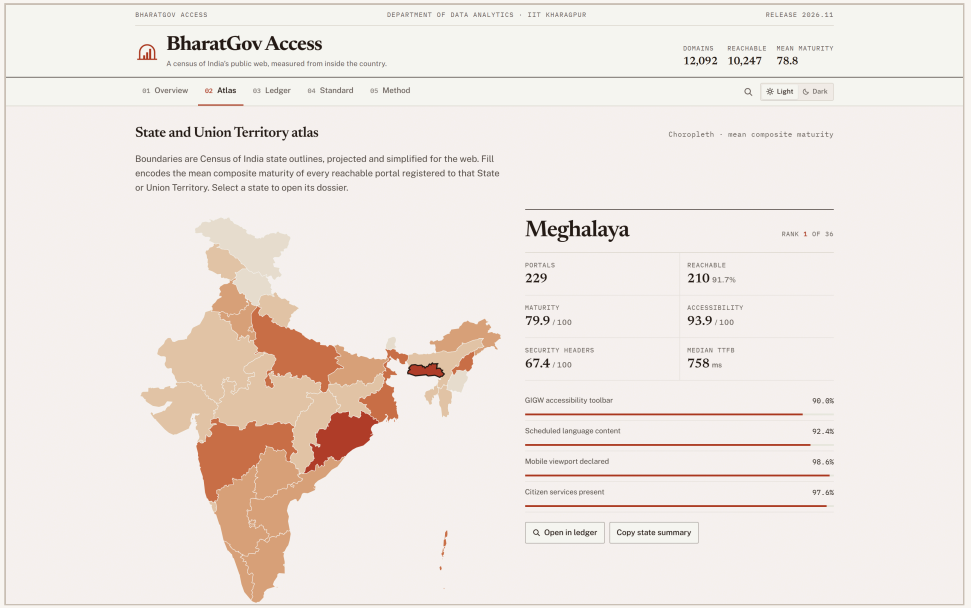
Overseas crawlers report a third of these portals as dead. They are not.

State data centres, NIC gateways and departmental reverse proxies drop foreign traffic at the perimeter. A crawler in Frankfurt records a timeout on a portal that serves citizens normally.

01 **Perimeter Geo-IP drops.** RailTel, NIC and state data centre firewalls silently terminate TCP handshakes from overseas cloud regions, producing false unreachability above 66%.

02 **Shallow samples.** Prior work evaluates 50–200 central ministry homepages, bypassing the district and municipal portals where most citizen transactions happen.

03 **Metric bloat.** Tools harvest 70+ uncalibrated variables and assign misleading zero scores to unreachable hosts instead of recording a diagnostic.



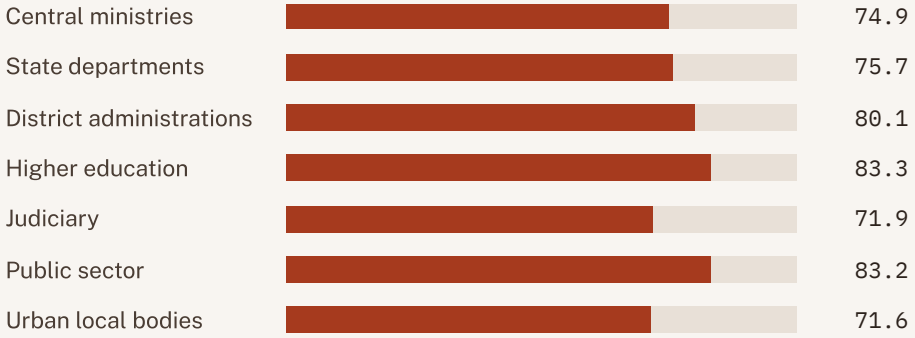
ATLAS · REAL CENSUS BOUNDARIES, CHOROPLETH BY MEAN MATURITY

The fix

Every measurement originates inside the domestic peering fabric. That single change moves observed reachability from roughly 33% to 84.7% and turns a timeout into a genuine signal.

Accessibility is solved. Security is not.

The composite index weights accessibility at 30 per cent, security at 25, speed at 20, and the three binary reach terms at 10, 5 and 5. District administrations lead on accessibility because they inherit NIC's S3WaaS template; higher education and public sector lead on security because they run their own edge.



Mean composite maturity by tier, reachable portals only.

TIER	PORTALS	LIVE	ACC.	SEC.	MATURITY
Central ministries	645	532	90.8	78.6	74.9
State departments	2,935	2,017	92.0	68.4	75.7
District administrations	7,468	6,716	95.0	65.8	80.1
Higher education	352	342	91.6	84.7	83.3
Judiciary	292	271	91.0	63.2	71.9
Public sector	260	247	91.8	84.4	83.2
Urban local bodies	111	101	90.9	63.7	71.6
National	**12,092**	**10,247**	**93.8**	**68.0**	**78.8**

The gap in one line

The tier with the highest accessibility (district administrations, 95.0) is not the tier with the highest security (higher education, 84.7). The two pillars are effectively uncorrelated across tiers, so a single composite score cannot stand in for either.

GIGW TOOLBAR
91.0%
Font and contrast controls

SCHEDULED LANGUAGE
86.0%
Eighth Schedule scripts

MOBILE VIEWPORT
97.9%
width=device-width

Three results that hold across the full population.

01 **The S3WaaS dividend is real but bounded.** S3WaaS reaches 95.2 mean accessibility and 100% toolbar penetration against 84.3 and 68% for legacy ASP.NET. It buys nothing on security: 66.8 against 85.1 for modern SPAs. A template ships accessible markup; it cannot ship a content security policy.

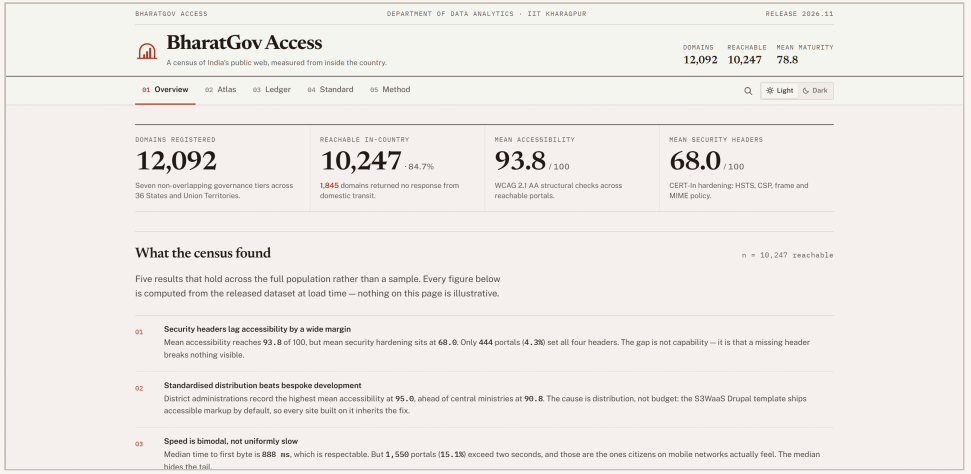
02 **Header hardening is the real deficit.** Transport encryption is universal at 100%, but only 444 portals (4.3%) set HSTS, CSP, frame options and MIME protection together. The mean header score is 68.0.

03 **Speed is bimodal, not uniformly slow.** Median TTFB is 888 ms, which is respectable, but 1,550 portals (15.1%) exceed two seconds. The median hides the tail citizens actually feel.

PLATFORM	PORTALS	SHARE	ACC.	SEC.	MATURITY
S3WaaS	6,480	63.2%	95.2	66.8	81.0
Custom CMS	2,671	26.1%	92.6	67.6	74.4
Modern SPA	651	6.4%	91.6	85.1	82.3
Legacy ASP.NET	399	3.9%	84.3	60.3	68.5
WordPress	**46**	**0.4%**	**90.8**	**74.5**	**70.8**

What this implies for policy

Accessibility can be fixed by distribution — one template, thousands of portals. Security cannot: it is an edge decision made per deployment. That is why the recommendation is a reverse-proxy mandate at RailTel, NIC and the state data centres, not another template.



OVERVIEW · EVERY FIGURE COMPUTED AT LOAD TIME

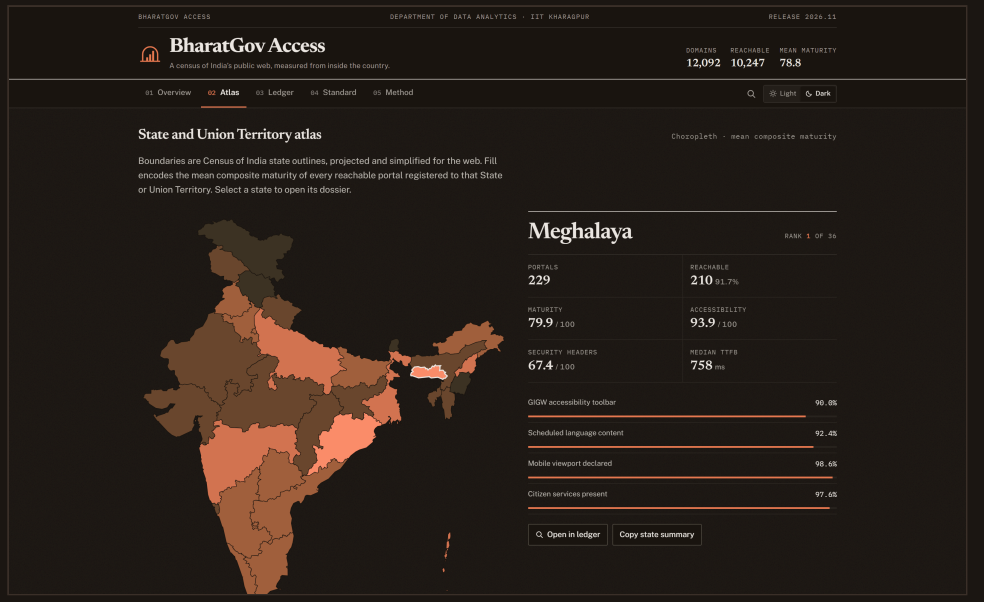
Everything is reproducible in one command.

INSTALL · QUERY · REPRODUCE

```
pip install bharatgov-access

import bharatgov_access as bga

bga.load()                    # 12,092 domains
bga.load(reachable=True)  # 10,247 live
bga.tier_summary()            # Table I
bga.architecture_summary()  # Table II
bga.state_ranking()           # Section V-C
```



ATLAS · DARK THEME

#	STATE / UT	PORTALS	MATURITY
1	Meghalaya	210	79.9
2	Odisha	252	79.8
3	Uttar Pradesh	389	79.6
4	Nagaland	210	79.5
5	Maharashtra	279	79.5

PACKAGE

465 KB wheel
dataset bundled
14 tests

OBSERVATORY

5 views
light and dark
keyboard-first

PAPER

IEEE two-column
tables verified
against release